

## QUANTITATIVE METHODS I · WEEK 1

# Equations, functions and graphs

Solving equations, what a function is, and the shapes you have to recognise on sight.

- 1.1 Linear equations and systems
- 1.2 Quadratic equations in business
- 1.3 What a function is
- 1.4 Domain and range
- 1.5 The families of function
- 1.6 Exponential functions
- 1.7 Logarithmic functions
- 1.8 Test yourself

## 1.1 Linear equations and systems

**First order equation.** One unknown, first power only.

$$ax + b = 0, \quad a \neq 0 \quad \Rightarrow \quad x = \frac{-b}{a}$$

Everything else is arithmetic to get it into that shape.

- 1 **Clear brackets, then fractions.** Multiply every term by the common denominator, the right-hand side included.
- 2 **Collect the unknown on one side and the numbers on the other.** Every term changes sign as it crosses the equals.
- 3 **Divide by the coefficient of the unknown.**
- 4 **For two unknowns you need two independent equations.** Independent means neither is a multiple of the other.
- 5 **Eliminate, or substitute.** Scale one equation so a letter cancels when you add. Or make one equation say  $x = \text{something}$  and put that into the *other* one.
- 6 **Check the pair in both equations.** One equation cannot catch an error.

**Solve  $3x + 1 = 5x + 7$**

$$3x - 5x = 7 - 1$$

Unknown left, numbers right. Both signs flip as they cross.

$$-2x = 6$$

Collect.

$$x = -3$$

Divide by  $-2$ . Check:  $3(-3) + 1 = -8$  and  $5(-3) + 7 = -8$ .

### Solve $2x + y = 18$ and $-x + 5y = 2$ , both ways

$$-2x + 10y = 4$$

By elimination: double the second equation.

$$11y = 22, \text{ so } y = 2$$

Add the two. The  $x$  terms cancel.

$$x = 5y - 2$$

By substitution instead: isolate  $x$  in the second.

$$2(5y - 2) + y = 18$$

Put it into the first.

$$x = 8, y = 2$$

Same answer, as it has to be.

**Three outcomes.** One solution. None, when the letters vanish and something false is left. Infinitely many, when the letters vanish and something always true is left. None of the three is a mistake.

#### WATCH OUT

- Multiplying an equation by something and forgetting its right-hand side.
- $-(x - 4) = -x + 4$ . The minus reaches both terms.
- Substituting back into the equation you just rearranged. It always gives  $0 = 0$ .

#### LEARN IT ON KHAN ACADEMY

##### **Video** First order equations · Why we do the same thing to both sides

Algebra 1, unit 2: Solving equations and inequalities

##### **Video** Systems, by substitution · Solving systems of equations with substitution

Algebra 1, unit 6: Systems of equations

##### **Article** The method, written out · Multi-step equations review

Algebra 1, unit 2: Solving equations and inequalities

#### PRACTISE IT ON KHAN ACADEMY

##### **Exercise** First order equations · Equations with variables on both sides

Algebra 1, unit 2: Solving equations and inequalities

##### **Exercise** Systems, by elimination · Systems of equations with elimination

Algebra 1, unit 6: Systems of equations

## 1.2 Quadratic equations in business

**Standard form.**

$$ax^2 + bx + c = 0, \quad a \neq 0$$

**The discriminant.**

$$\Delta = b^2 - 4ac$$

It answers how many roots there are before you find any:  $\Delta > 0$  two,  $\Delta = 0$  one,  $\Delta < 0$  none among the real numbers.

- 1 **Move everything to one side.** Nothing works until the right-hand side is zero.
- 2 **Read off a, b and c with their signs.** The signs are where the marks go.
- 3 **Work out the discriminant.** It tells you what to expect, and whether the roots will be tidy.
- 4 **Factor if it factors. Otherwise use the formula.**

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

- 5 **For a business question, set revenue equal to cost.**  $R(q) = C(q)$  rearranges to a quadratic, and break-even is its root.
- 6 **Discard any root the situation forbids.** A negative quantity or a negative price is not an answer.

**Solve  $3x^2 + 3x - 18 = 0$**

$$\Delta = 3^2 - 4(3)(-18) = 225$$

Positive, and a perfect square, so two tidy roots.

$$x = \frac{-3 \pm 15}{6}$$

Into the formula.

$$x = 2 \quad \text{or} \quad x = -3$$

It also factors:  $3(x - 2)(x + 3) = 0$ , which is faster when you spot it.

**Break-even: cost  $C(q) = 6q + 80\,000$ , revenue  $R(q) = 46q$**

$$46q = 6q + 80\,000$$

Break-even is where the two are equal.

$$40q = 80\,000$$

Collect.

**q = 2 000 units**

Below that the firm loses money, above it makes money.

#### WATCH OUT

- $b = -7$  gives  $b^2 = 49$ . Squaring kills the sign.
- $(x - 2)(x + 5) = 4$  tells you nothing. Only a product equal to *zero* splits.
- A negative  $\Delta$  in a business question usually means the target is never reached.

#### LEARN IT ON KHAN ACADEMY

##### Video The formula · The quadratic formula

Algebra 1, unit 14: Quadratic functions and equations

##### Video What the discriminant tells you · Using the quadratic formula: number of solutions

Algebra 1, unit 14: Quadratic functions and equations

##### Article The discriminant · Discriminant review

Algebra 1, unit 14: Quadratic functions and equations

#### PRACTISE IT ON KHAN ACADEMY

##### Exercise Using the formula · Quadratic formula

Algebra 1, unit 14: Quadratic functions and equations

##### Exercise Reading the discriminant · Number of solutions of quadratic equations

Algebra 1, unit 14: Quadratic functions and equations

## 1.3 What a function is

**Function.** A rule that gives each allowed input **exactly one** output. That is the whole definition.

**f(x).** The output at x. It is not f multiplied by x.

**Four representations.** A formula, a table, a graph, a sentence. The same function, four ways, and you are expected to move between them. A table is exact and sparse; a graph shows the shape and hides the decimals.

**Vertical line test.** If a vertical line meets the graph twice, it is not a function: that input would have two outputs.

- 1 **To evaluate, replace every x by the whole input.** Brackets included.  $f(a + 1)$  needs  $(a + 1)$  everywhere x appeared.
- 2 **Read a table exactly and a graph approximately.**
- 3 **Slope is rise over run, and it is the same wherever you measure it.**

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

- 4 **To sketch a line, plot two points and join them.** The intercepts are usually the easiest two.
- 5 **A vertical line has no slope.**  $x = k$  is not a function of  $x$ , and its slope is undefined rather than zero.

**$f(x) = x^2 - 3x$ , evaluated at  $a + 1$**

$$(a + 1)^2 - 3(a + 1)$$

Replace every  $x$  by the whole of  $(a + 1)$ .

$$= a^2 + 2a + 1 - 3a - 3$$

Expand both.

$$= a^2 - a - 2$$

Collect.

**The plane.** Two axes crossing at the origin. A point is an ordered pair: along first, then up. The horizontal axis carries the input, the vertical the output, and that order is a convention nobody breaks.

- 1 **To graph by hand, make a table first.** Choose inputs, work out each output, plot the pairs, join them.
- 2 **Choose inputs that show the shape.** Include zero and either side of anything interesting.
- 3 **To graph in a spreadsheet, put the inputs in one column and the outputs in the next.** Then insert a scatter chart with smooth or straight lines, not a bar chart.
- 4 **Label both axes with what they are and what unit they are in.** A graph without units is a picture, not evidence. Give it a title too.
- 5 **Choose a scale that uses the space.** Starting every axis at zero can squash the thing you are trying to show.

#### WATCH OUT

- $f(a + 1)$  is not  $f(a) + 1$ . The input goes in whole.
- Reading a value off a sketch when the question wanted it solved for.
- Slope is  $\frac{\Delta y}{\Delta x}$ . Subtract the coordinates in the same order top and bottom.

#### LEARN IT ON KHAN ACADEMY

##### **Video** What counts as a function · Recognizing functions

Algebra 1, unit 8: Functions

##### **Video** Evaluating a function · Worked example: Evaluating functions from equation

Algebra 1, unit 8: Functions

##### **Video** Lines, slope and intercept · Worked examples: slope-intercept intro

Algebra 1, unit 5: Forms of linear equations

## PRACTISE IT ON KHAN ACADEMY

**Exercise Evaluating · Evaluate functions**

Algebra 1, unit 8: Functions

**Exercise The vertical line test · Recognize functions from graphs**

Algebra 1, unit 8: Functions

**Exercise Reading a line off a graph · Slope-intercept equation from graph**

Algebra 1, unit 5: Forms of linear equations

## 1.4 Domain and range

**Domain.** Every input the rule accepts.**Range.** Every output that actually occurs. Usually smaller than you expect.

- 1 **Assume every real number is allowed, then take away what is forbidden.**
- 2 **A denominator may not be zero.** Set it to zero, solve, and exclude those values.
- 3 **An even root may not have a negative underneath.**  $\sqrt{x - 3}$  needs  $x \geq 3$ .
- 4 **A logarithm needs a strictly positive argument.**
- 5 **Keep whatever survives all of them.**
- 6 **Write it however you like.** An inequality, a set, or an interval. All three say the same thing.
- 7 **From a graph, read both off the axes.** The domain is how far the curve reaches left and right, the range how far it reaches up and down.

**Find the domain of  $f(x) = \frac{x}{x^2 - x - 2}$**

$$x^2 - x - 2 = 0$$

The only thing that can go wrong is a zero denominator.

$$(x - 2)(x + 1) = 0$$

Factor it.

$$x \neq 2 \text{ and } x \neq -1$$

The domain is every real number except those two.

**WATCH OUT**

- Solving the denominator and then reporting those values as the domain. They are what is excluded.
- Checking the denominator and forgetting a root or a logarithm elsewhere in the same expression.
- Giving the domain when the question asked for the range.

## LEARN IT ON KHAN ACADEMY

**Video** What is the domain of a function?

Algebra 1, unit 8: Functions

**Video** What is the range of a function?

Algebra 1, unit 8: Functions

## PRACTISE IT ON KHAN ACADEMY

**Exercise** Finding a domain · Determine the domain of functions

Algebra 1, unit 8: Functions

**Exercise** Reading both off a graph · Domain and range from graph

Algebra 1, unit 8: Functions

## 1.5 The families of function

**The seven.** Constant, linear, power, quadratic, polynomial, rational, absolute value. Naming the family tells you the shape before you draw anything, which is what the rest of the course leans on.

**Power function.**

$$f(x) = x^m$$

The exponent decides the shape.  $m = 1$  a line,  $m = 2$  a parabola,  $m = 3$  a curve through the origin that keeps its sign,  $m = -1$  a hyperbola with the axes as asymptotes. An even  $m$  is symmetric about the vertical axis; an odd  $m$  about the origin.

**Absolute value.**

$$|x| = x \text{ if } x \geq 0, \quad -x \text{ if } x < 0$$

Distance from zero, so never negative. Its graph is a V at the origin.

**Case-defined.** Several rules over several intervals. Common in pricing: one rate up to a threshold, another above it.

- 1 **Look at the formula, not at the picture.** The form is what names the family.
- 2 **Constant: no  $x$  at all.**  $h(x) = 2$  returns 2 whatever you give it.
- 3 **Linear: one  $x$ , first power.** A straight line.
- 4 **Power:  $x$  raised to a fixed exponent.** The exponent is the whole story of the shape.
- 5 **Quadratic: there is an  $x^2$ .** A parabola, opening upwards when  $a > 0$ .
- 6 **Rational: an  $x$  in a denominator.** So find the domain before anything else.

- 7 **Case-defined: graph each piece only over its own interval.** And check what happens at the joins, which is where the marks are.

**A delivery charge: 8 CHF under 20 kg, free from 20 kg**

$$C(w) = 8 \text{ for } 0 < w < 20$$

First piece, over its own interval only.

$$C(w) = 0 \text{ for } w \geq 20$$

Second piece.

**A step at  $w = 20$**

The join is the interesting point, and the one questions ask about.

**WATCH OUT**

- $|x|$  is not always  $x$ . It is  $-x$  whenever  $x$  is negative.
- Drawing a case-defined function as one continuous curve. It usually jumps.
- Naming the family from the graph rather than the formula, and getting it the wrong way round.

**LEARN IT ON KHAN ACADEMY**

**Video Parabolic functions · Intro to parabolas**

Algebra 1, unit 14: Quadratic functions and equations

**Video Graphs of absolute value functions**

Algebra 1, unit 10: Absolute value and piecewise functions

**Video Graphs of rational functions**

Precalculus, unit 4: Rational functions

**PRACTISE IT ON KHAN ACADEMY**

**Exercise Parabolas intro**

Algebra 1, unit 14: Quadratic functions and equations

**Exercise Case-defined functions · Evaluate piecewise functions**

Algebra 1, unit 10: Absolute value and piecewise functions

## 1.6 Exponential functions

**Exponential function.**

$$f(x) = b^x, \quad b > 0, \quad b \neq 1$$

Growth when  $b > 1$ , decay when  $b$  is between 0 and 1.

**Compound interest.**

$$S = P(1 + r)^n$$

P the principal, r the rate per period as a decimal, n the number of periods.

**The number e.** The limit of  $(1 + \frac{1}{n})^n$  as n grows, about 2.718.  $\exp(x)$  and  $e^x$  are the same thing.

- 1 **A quantity that multiplies by a fixed factor each step is exponential.** One that adds a fixed amount each step is linear. That distinction is most of the exam questions.
- 2 **Growing by p% each period multiplies by the same number each period.**

$$\times (1 + \frac{p}{100})$$

- 3 **Read the period off the question.** 8% a month and 8% a year are not the same number.
- 4 **Put the rate in as a decimal.** 2% is 0.02, so the factor is 1.02.
- 5 **Do not round until the end.** Rounding at each compounding moves the answer.

**10 000 CHF at 2% a year, compounded annually, for three years**

$$S = 10\,000(1.02)^3$$

Straight into the formula.

$$= 10\,612.08 \text{ CHF}$$

To the franc, 10 612.

**500 subscribers growing 8% a month, after 12 months**

$$S = 500(1.08)^{12}$$

The period is a month, so n counts months.

$$\approx 1\,259$$

Two and a half times the start, from 8% a month.

**WATCH OUT**

- Using the annual rate with a monthly period, or the other way round.
- $(1 + r)^n$  is not  $1 + r^n$ .
- Treating steady growth as linear. Over twelve periods the gap is enormous.

LEARN IT ON KHAN ACADEMY

**Video What makes it exponential · Exponential vs. linear growth**

Algebra 1, unit 12: Exponential growth and decay

**Video Where e comes from · e and compound interest**

Algebra 2, unit 8: Logarithms

**Video e as a limit · e as a limit**

Algebra 2, unit 8: Logarithms

PRACTISE IT ON KHAN ACADEMY

**Exercise Growth and decay in context · Exponential expressions word problems (numerical)**

Algebra 1, unit 12: Exponential growth and decay

**Exercise The two shapes · Graphs of exponential growth**

Algebra 1, unit 12: Exponential growth and decay

**1.7 Logarithmic functions****Logarithm.**

$$y = \log_b x \quad \text{if and only if} \quad b^y = x$$

It answers: to what power do I raise the base to get this number?

**The two on your calculator.** log is base 10. ln is base e.**Each undoes the other.**

$$\log_b(b^x) = x \quad b^{\log_b x} = x$$

- 1 To convert, ask what power the base is raised to.** Reading the statement in the other form is most of the work.
- 2 Without a calculator, write the argument as a power of the base.**  $\log_2 32$ : 32 is  $2^5$ , so the answer is 5.
- 3 Only a strictly positive argument has a logarithm.** Check that before anything else.
- 4 To solve an equation with the unknown in the exponent, take the log of both sides.** The power rule then brings it down where you can reach it.
- 5 Use ln for anything involving e, and log for anything involving powers of ten.**

**Evaluate without a calculator**

$$\log 100 = \log_{10}(10^2) = 2$$

Write the argument as a power of the base.

$$\ln e^3 = 3$$

The natural log undoes e.

$$\log_3 81 = 4$$

Because  $3^4 = 81$ .

### How long at 5% a year to double your money?

$$(1.05)^n = 2$$

Double means the factor reaches 2.

$$n \ln 1.05 = \ln 2$$

Take logs, then the power rule brings n down.

$$n = \frac{\ln 2}{\ln 1.05} \approx 14.2 \text{ years}$$

So fifteen full years.

**The shape of the curve.** For  $b > 1$ ,  $y = \log_b x$  passes through  $(1, 0)$ , rises for ever but ever more slowly, and falls away without limit as  $x$  approaches zero from the right. The vertical axis is an asymptote it never touches, so the domain is  $x > 0$  and the range is all of  $\mathbb{R}$ . It is the reflection of  $y = b^x$  in the line  $y = x$ , which is what being its inverse looks like.

#### WATCH OUT

- $\log 0$  and the log of a negative number do not exist.
- $\log(M + N)$  does not simplify. There is no rule for a sum inside a logarithm.
- $\log$  on its own means base 10 here, but base  $e$  in some software. Check which you are using.

#### LEARN IT ON KHAN ACADEMY

##### Video What a logarithm is · Introduction to logarithms

Algebra 2, unit 8: Logarithms

##### Video The shape of the graph · Graphs of logarithmic functions

Algebra 2, unit 9: Transformations of functions

##### Article The definition, written out · Intro to logarithms

Algebra 2, unit 8: Logarithms

#### PRACTISE IT ON KHAN ACADEMY

##### Exercise Converting between the two forms · Relationship between exponentials & logarithms

Algebra 2, unit 8: Logarithms

##### Exercise Evaluating without a calculator · Evaluate logarithms

Algebra 2, unit 8: Logarithms

##### Exercise Graphing · Graphs of logarithmic functions

Algebra 2, unit 9: Transformations of functions

## 1.8 Test yourself

Do this week's homework on the course platform, then check it against the answers when they are posted. That is the test. Everything above is how to be ready for it.

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Student notes, not official course material.